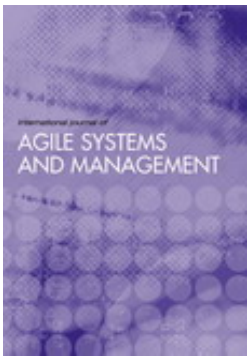
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R. Chadwick Holmes, Daniel Hussmo

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Architectural strategies for agentic AI in circular ecosystem formation and orchestration

R. Chadwick Holmes*

Exploration Excellence,
Chevron USA, Inc.,
Houston, TX 77002, USA
Email: chadwick.holmes@chevron.com
*Corresponding author

Daniel Hussmo

Saab Avionics,
Saab AB,
Huskvarna, 561 39, Sweden

and

School of Engineering,
Jönköping University,
Jönköping, 553 18, Sweden
Email: daniel.hussmo@saabgroup.com

Abstract: A key enabler for a sustainable and resilient future is the deployment of a circular economy, where closing material loops enhances resource efficiency. Collaboration within circular ecosystems at the meso level allows companies to jointly reduce and reuse materials, yet identifying suitable partners and adapting to dynamic orchestration challenges remains difficult. Recent advances in agentic artificial intelligence (AI) offer promising support, although questions persist regarding optimal architectures and practical implementation. This paper presents an agentic AI framework to support circular ecosystem formation and orchestration, analysing agent complexity, coordination, and adoptability. Results show that architectures with more specialised agents and focused task allocations achieve higher adoptability, promoting transparency and user acceptance. Additionally, workflow artefacts serve as boundary objects, facilitating mutual understanding and effective decision-making among ecosystem participants. These findings highlight the importance of intentional design for system impact, informing future research and the development of effective AI-enabled circular ecosystem strategies.

Keywords: agentic artificial intelligence; circularity; circular ecosystems; sociotechnical systems; joint ventures; system architecture; complexity; adoptability; boundary objects; decision support.

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Biographical notes: R. Chadwick Holmes is a Senior Manager for Exploration Technology, Data & AI at Chevron. He holds a PhD in Marine Geophysics and a MS in Engineering and Management. His research interests include integrated exploration geoscience methods, systems engineering for value and impact, and applications of AI for solving complex sociotechnical problems.

Daniel Hussmo is a Product Manager at Saab Avionics and is an affiliated researcher at the School of Engineering at Jönköping University. He holds a licentiate degree in Production Systems and an MS in Industrial Engineering and Management. His research interests include circular ecosystems, circularity for defence and aerospace industry, and AI-applications for creating agency in sociotechnical systems.

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1 Introduction

A key enabler for a sustainable and resilient future is a circular economy, which functions by slowing and closing material loops (Kirchherr et al., 2017). Central to this concept is a holistic view that shifts from examining single actors at a micro level, to examining intersections between actors on a meso level and considering macro-scale implications of policy. An essential aspect at the meso level is the deployment of circular ecosystems, which have been characterised in different ways by recent studies (Aryee et al., 2025; Geissdoerfer et al., 2025; Pietrulla, 2022), introducing concepts like circular innovation systems and knowledge systems (Konietzko et al., 2020). This study adopts the definition from Geissdoerfer et al. (2025):

> "*Circular economy ecosystems describe systems of stakeholders (producers, suppliers, etc.), their different roles (orchestrators, brokers, etc.), and the interactions and interrelations (knowledge and information, material and energy, and economic flows) between them, focused at extending, cycling, intensifying, and dematerialising resource loops.*"

Geissdoerfer et al.'s framework emphasises three main points: who the stakeholders are, how they interact, and how these actions connect to resource loops.

Firms implementing circular ecosystem practices must therefore consider the roles and resource flows of different stakeholders as part of an effective strategy. Chirumalla et al. (2025) suggest examining ecosystems from various scales, including firm-level evaluation, ecosystem formation, orchestration, and adaptation at the organisation level. From the standpoint of the firm, both ecosystem formation and orchestration present distinct challenges. Identifying appropriate stakeholders is often resource-intensive, as suitable

partners may not be readily apparent. Because ecosystems must benefit all participants, volatility from policy shifts, trade disputes, and supply chain disruptions requires firms to maintain agility in both forming and orchestrating ecosystems (Aryee et al., 2025).

These challenges raise the question of how circular ecosystems can be established and managed with agility. One promising approach leverages artificial intelligence (AI), which has been identified as a key enabler in the circular economy (Rodrigues Dias et al., 2022). While existing research has explored the intersection of AI and circularity from both logistics (Wilson et al., 2022) and product perspectives, methodological developments addressing the circular ecosystems perspective remain underexplored (Tutore et al., 2024). Recent work by Holmes and Hussmo (2025) has demonstrated that agentic AI can facilitate the formation of these ecosystems by identifying collaborative venture opportunities among a predefined set of firms. Given the nascent stage of this field, defining robust AI architectures for both forming and orchestrating circular ecosystems is crucial for advancing both AI applications and circular economy research.

Engineering an agentic AI approach to address complex challenges like joint venture optimisation for circular ecosystem development closely parallels the structured decision-making process found in standard engineering design. In scenarios where prior design analysis is lacking, engineers might be susceptible to familiarity bias, defaulting to known solutions and personal preferences (Viswanathan and Linsey, 2020). This tendency can lead to systems that are not easily scalable, exhibit suboptimal performance, or demonstrate unpredictable emergent behaviours that undermine user trust and adoption (Gero and Kannengiesser, 2004; Sapkota et al., 2026).

The successful incorporation of agentic AI into problem-solving and decision-making strongly depends on user acceptance, which can be influenced by the design of multi-agent systems. Prior research on traditional multi-agent systems established that complex emergent behaviours can arise when agents conform to simple rule-based actions (Schelling, 1971; Reynolds, 1987; Ferber and Weiss, 1999). The likelihood of unexpected outcomes stands to increase with generative AI agents using large language models (LLMs) and probabilistic reasoning to make decisions (Sapkota et al., 2026). Human factors also play a key role; when AI systems appear inexplicable, diminished trust can result in users dismissing or disregarding their outputs (ECHLEGAI, 2020). The practical value of agentic AI is therefore fundamentally dependent on engineering design choices that support user adoption. Specifically, design decisions that improve system transparency and enhance traceability directly promote user acceptance by making the system more comprehensible and trustworthy (ECHLEGAI, 2020; Glikson and Woolley, 2020; Kizilcec, 2016).

Addressing these critical dependencies, this study focuses on designing and assessing agentic AI architectures that intentionally foster circular ecosystems through a design-oriented, pre-deployment evaluation. The central objective is to establish an architectural framework prior to actual system development. Such a framework guides design choices – including agent specialisation, artefact management, and coordination structures – that increase system transparency and traceability. These attributes, in turn, promote user acceptance, which is essential for systems to achieve their intended circular ecosystem impact. Recognising the methodological gap in existing literature at the intersection of agentic AI and circular ecosystems, we investigate the potential and limitations of agentic AI, both architecturally and organisationally, as a catalyst for forming and orchestrating these ecosystems. The subsequent methodology provides a transparent and rigorous framework for architectural design and assessment, using circular joint venture analysis as a practical demonstration.

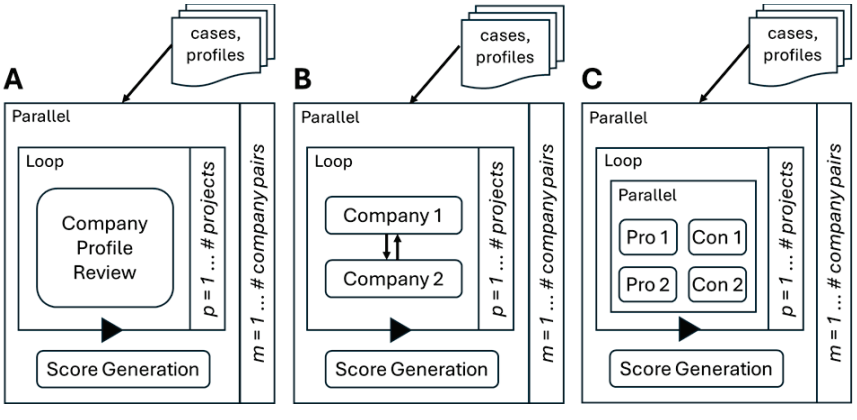
2 Methodology

2.1 Work breakdown structure

To conduct a thorough architectural analysis of an agentic AI system, the intended agentic workflow must first be restructured into a work breakdown structure (WBS), treating each step as an independent action. For greater clarity, these actions can be modelled using the object process model (OPM) framework, which identifies a singular object, process, and operand (International Organization for Standardization, 2024). Critically, repetitive steps must be listed separately rather than consolidated to facilitate grouping of similar actions by the process. For instance, rather than stating 'Navigate to the websites for companies A through C', the WBS requires distinct steps: 'Step 1: navigate to company A's website', 'Step 2: navigate to company B's website', 'Step 3: navigate to company C's website'.

For this study, Holmes and Hussmo (2025) circular joint venture analysis workflow (Figure 1(A)) was extracted as a preliminary WBS using generative AI (OpenAI, 2025), then expanded such that each step represented an atomic instruction. This expansion was iteratively refined through author review until every instruction reached a sufficient level of granularity for further analysis.

Figure 1 (A) (Design 1) Simplified schematic of the circular ecosystem joint venture analysis workflow from (Holmes and Hussmo, 2025). Note the company analysis is performed via an assessment of curated company profiles; (B) (Design 2) Variation on the same workflow with AI agents as company representatives directly negotiating through conversation and (C) (Design 3) Third architecture using a pro/con analysis from different company perspectives for a comprehensive evaluation



Several aspects of the workflow described in (Holmes and Hussmo, 2025) were modified for this analysis. The number of circularity projects evaluated was reduced from 10 to 5, and the number of companies considered was reduced from 20 to 3. Company selection was treated as a workflow precondition and not addressed in the WBS. Data sources for company information were broadened to include corporate annual reports, sustainability reports, SEC filing equivalents (10K or 20F), and press releases from the past 5 years. The final version of the WBS included 174 steps for workflow Design 1 (Table 1).

Figure 1 also illustrates alternative workflow designs for agentic joint venture analysis. In Design 2 (189 steps), AI agents act as company representatives who negotiate potential

Table 1 Representative subset of the Design 1 work breakdown structure (WBS), also illustrating the key artefact associated with each WBS step

ID	*WBS*	*Artefact*
1	Define scoring rubric for Reduce (scale 1–5)	Scoring Rubric
2	Define scoring rubric for Reuse (scale 1–5)	Scoring Rubric
3	Define scoring rubric for Recycle (scale 1–5)	Scoring Rubric
4	Define scoring rubric for Strategic Alignment (scale 1–5)	Scoring Rubric
5	Define output format for profiles	Output Format
6	Define output format for scenario evaluations	Output Format
7	Define output format for scorecards	Output Format
8	Search web for sustainability trends in relevant industries	Themes and Trends
12	Generate Project_1	Project Ideas
13	Generate Project_2	Project Ideas
...	...	...
17	Search Company_A Annual Report	Annual Report
18	Download Company_A Annual Report	Annual Report
19	Search Company_A Sustainability Report	Sustainability Report
20	Download Company_A Sustainability Report	Sustainability Report
21	Search Company_A Regulatory Filing	Regulatory Filing
22	Download Company_A Regulatory Filing	Regulatory Filing
...	...	...
173	Highlight highest scoring company pairs in final report	Final Report
174	Highlight associated project ideas in final report	Final Report

circular economy partnerships directly with one another, evaluating project feasibility and compatibility from their respective company perspectives (Figure 1(B)). Design 3 (219 steps) evaluates company compatibility by weighing both advantages ('pro view') and disadvantages ('con view'), balancing the potential for optimistic bias with critical scrutiny (Figure 1(C)). All three designs were analysed using the same approach after WBS decomposition.

2.2 Embeddings and similarity matrix

The WBS steps were enriched using a generative AI approach (Holmes and Kumar, 2023) to provide consistent contextual detail for every workflow instruction. An LLM first rephrased each task into a two- to three-sentence expansion, standardising the amount of context provided. The LLM then reformulated this expanded description into a succinct action statement that precisely specified the required actions and any tools necessary for a generative AI agent to accomplish the task. To enhance clarity and workflow organisation, each action was further decomposed into object-process-operand components, aligning with the OPM standard (International Organization for Standardization, 2024). All generated representations for each workflow step were stored within the WBS table for further analysis.

LLMs operate by converting text into a condensed form known as an embedding vector (Reimers and Gurevych, 2019). This representation allows for the quantification of contextual similarity between related workflow steps using analytical measures such as cosine similarity. This study utilised OpenAI's text-embedding-3-small model to produce 1536-dimensional vectors per workflow step, which were also saved to the WBS table.

General-purpose embedding models like text-embedding-3-small produce baseline similarity across workflow steps due to consistent context. To address this, Holmes and Kumar (2023) developed a thresholding technique that uses the cumulative distribution function (CDF) to define a cut-off point, setting similarity values below this point to zero. This cut-off is identified at the 'knee' of the CDF and then shifted 20% to the left to prevent excessive truncation. Rather than converting the remaining non-zero values to 1.0 to maximise contrast, min-max normalisation was applied to preserve the variability of the resulting matrix.

2.3 Task clustering

Following matrix conditioning, the similarity matrix was reorganised to concentrate higher similarity values along the diagonal while diminishing those off-diagonal. This reordering facilitated the clustering of actions with strong pairwise similarities, identifying groups of tasks that could feasibly be managed by a single generative AI agent.

To objectively identify the optimal number of clusters within this matrix, a composite approach using multiple cluster evaluation metrics was employed. The Silhouette score offers a local assessment of cluster cohesion, quantifying the alignment of a data point to its assigned cluster relative to others, where positive scores indicate robust association and negative values suggest potential misclassification (Rousseeuw, 1987). The Calinski-Harabasz index evaluates overall clustering quality by computing the ratio of cluster separation to compactness, with higher values signifying better clustering (Caliński and Harabasz, 1974). The Davies-Bouldin index quantifies worst-case inter-cluster similarity by averaging the maximum ratio of within-cluster scatter to between-cluster separation across all cluster pairs (Davies and Bouldin, 1979). Since lower Davies-Bouldin scores are desirable, minimising this metric is preferred. Additionally, Spectral Clustering Analysis calculates Laplacian eigenvalues from the similarity matrix and identifies significant changes in successive eigenvalue magnitudes (Von Luxburg, 2007). These changes suggest suitable cluster counts, as detailed in (Holmes and Kumar, 2023).

Each metric was normalised using min-max scaling to ensure values fall between zero and one. Davies-Bouldin scores were inverted during normalisation to reflect the minimisation criterion. These four normalised scores were then averaged to create a single composite score. Maxima indicate optimal cluster counts in the data. While the highest peak suggests a primary candidate for cluster count, secondary peaks can also be considered valid alternatives, allowing some flexibility in cluster selection.

2.4 Complexity analysis

2.4.1 Agent complexity

A generative AI agent relies on a few core elements: an LLM for reasoning and decisions, an instructional prompt establishing its role, specific goals or tasks, and, optionally, a set of functions it can use as 'tools' to support its activities. Here, the tasks for each agent are outlined by the WBS steps allocated to each cluster, which effectively define the agent's role that should be described in its instructional prompt. Each task requires either inherent LLM reasoning or additional tools for completion. Consequently, the breadth of tools available to an agent not only reflects its capabilities but also adds complexity to the decision-making process regarding tool selection in various scenarios. More tools offer more choices, yet they also create a greater cognitive load for users attempting to understand agent behaviour.

To determine the appropriate tool list for each potential agent, generative AI (OpenAI, 2025) recommended agentic tools required to perform each WBS activity within each cluster. These recommendations were reviewed and validated by the authors to ensure accuracy and consistency. Equation (1) shows the complexity of a single agent as the count of unique tools for its associated tasks. System-level complexity is bounded by the maximum complexity among all individual agents:

$$c_{\text{agent}} = \max_{a_i \in \mathcal{A}} |T_i| \,, \tag{1}$$

where c_{agent} is the agent complexity score, $\mathcal{A}$ is the set of unique agents in the system, and $|T_i|$ is the cardinality of tools assigned to the ith agent (a_i).

2.4.2 Coordination complexity

In multi-agent systems, coordination complexity arises when agents rely on shared objects or artefacts. These dependencies require explicit coordination among agents to ensure consistent access and management of artefacts. Artefact-sharing creates stronger links between agents, increasing the likelihood that changes made by one agent impact others. Heightened interconnectedness can produce longer lead times, greater potential for rework, and higher cognitive demands on users trying to understand agent interactions. As the number of shared artefacts grows, the entanglement among agents intensifies, further amplifying coordination complexity within the system.

Coordination complexity scores were determined by mapping agents to artefacts in an Agent-Artefact (AA) matrix, where rows denote agents and columns represent artefacts. Workstep objects derived from OPM decomposition of WBS tasks formed the artefact list, which the authors reviewed to ensure consistent naming standards. The AA matrix is binary: a value of one indicates an agent-artefact interaction, while zeros populate all other cells. Once established, the Agent Coordination (AC) matrix is calculated through matrix multiplication, quantifying the number of shared artefacts between each pair of agents (equation (2)):

$$AC = AA \times AA^T. \tag{2}$$

In this representation, the matrix diagonal captures the number of artefacts each agent engages with, while off-diagonals show the number of shared artefacts between agents. The bigger the off-diagonal sum, the greater the coordination complexity score:

$$c_{multiagent} = \left(\sum_{i,k} AC_{ik} \right) - \text{Tr}(AC), \tag{3}$$

where Tr is the matrix trace, i.e., the sum of the diagonal entries.

2.4.3 Adoptability score

As sociotechnical systems, agentic systems only realise their full value when end users actually adopt them. Systems engineering practices typically rely on user testing and feedback to find the optimal balance between workflow abstraction and human-in-the-loop involvement for maximising both efficiency and user trust in the system (INCOSE, 2023). However, early-stage architectural design can precede system prototyping, and with no system embodiment for stakeholders to evaluate, alternative metrics must be used to judge the likely adoptability of the system.

Complex AI systems can be branded as inexplicable 'black boxes', reducing the willingness of users to accept their results (ECHLEGAI, 2020). Technology trust increases with transparency of system behaviour (Glikson and Woolley, 2020), suggesting that the more difficult the system is to comprehend, the lower the likelihood that potential users will rely on the system for problem solving or decision making (Kizilcec, 2016). Cognitive load theory describes understanding as a mixture of familiarity in long-term memory and the ability to hold many interconnected elements in working memory (Sweller et al., 2019). Assuming a user is unfamiliar with the agent-artefact connections within a multi-agent system, i.e., long-term memory does not come into play, the cognitive load imposed on the user (L) can be estimated by how complex it is for the user to understand the role of each agent. This user-focused cognitive load increases when an agent interacts with many artefacts and when those artefacts are shared with multiple other agents, requiring the user to comprehend these interconnections:

$$L = \sum_{j} AA_{ij} \times \left(1 + \sum_{k \neq i} AC_{ik}\right), \tag{4}$$

where the first sum defines the number of artefacts assigned to the ith agent, and the parenthesised terms define the shared connectivity with other agents.

Beyond cognitive load, additional factors contribute to a complete treatment of agentic AI system adoptability. Workflow complexity (S_i) measures the percentage of workflow steps allocated to each individual agent. Smaller workflow fractions equate to fewer worksteps per agent, clarifying agent-task associations and improving system comprehensibility. Tool complexity (T_i), the number of unique tools per agent, was already introduced in equation (1). Internal cohesion (I_i) captures agent self-containment by measuring the ratio of exclusive artefacts to all artefacts, indicating the degree of task independence. Conversely, external cohesion E quantifies external coupling through shared artefacts, which ideally should be minimised. E represents a single agent perspective on the full $c_{multiagent}$ coordination complexity score.

All factors in this adoptability framework reflect aspects of agent and system design that influence user comprehension and acceptance. Since each factor describes attributes of agents and their interactions, together they collectively characterise how difficult it is for users to understand, trust, and adopt the system. Equation (5) shows how these metrics can be consolidated into a single score:

$$\begin{aligned} a_{system} &= \min_{i \in \mathcal{A}} \left(a_{agent_i}\right) \\ &= \min_{i \in \mathcal{A}} \left(w_1\, \hat{S}_i{}' + w_2\, \hat{T}_i{}' + w_3\, I_i{}' + w_4\, \hat{E}_i{}' + w_5\right), \end{aligned} \tag{5}$$

where each w is a weight such that $\sum_i w_i = 1.0$, and x' is the min-max normalised value of metric x. Adoptability follows a maximisation objective, so metrics intended to be minimised must be inverted during min-max normalisation, indicated by $\hat{x}$. Each agent receives an adoptability score, and since system adoptability is most strongly influenced by the lowest trust level, the minimum score represents the overall system.

Equal weighting of the factors in this study (i.e., $w_i = 0.2$) represents an illustrative design choice, assuming each factor contributes proportionally to adoptability. However, alternative weighting schemes may be more appropriate depending on organisational context, decision-making priorities, or user populations. Future research could empirically validate optimal weighting schemes for specific circular ecosystem formation scenarios.

2.5 Methodological considerations

This architectural analysis relies on LLM assistance at multiple stages, including WBS expansion (Section 2.1), vector embedding generation (Section 2.2), and tool assignment to agents (Section 2.4.1). With appropriate guidelines and guardrails, generative AI offers both consistency and scalability for natural language tasks, but it also raises considerations warranting discussion. First, the quality of WBS expansion and workstep reframing depends on how the LLM interprets the original workflow, which can vary across different AI models or prompting strategies. To mitigate this, results were iteratively refined with author review at each stage. Second, embeddings were generated using a general-purpose model (text-embedding-3-small) that may not capture domain-specific nuances as effectively as fine-tuned alternatives, although the thresholding approach (Section 2.2) addresses baseline similarity issues. Third, tool assignments were validated to ensure accuracy and consistency, although alternative approaches (e.g., expert panels or LLM ensembles) could provide additional robustness checks in real-world applications. These choices prioritise transparency and reproducibility while acknowledging that different LLM selections or validation techniques may yield different clustering outcomes, similar to how multiple human experts would produce varying results when tackling the same problem without AI. Nevertheless, the overall architectural principles and assessment framework remain robust to these variations, as demonstrated by consistent trends in the results.

3 Results

Figure 2(A) displays the cosine similarity matrix for WBS task embeddings from the 174-step Design 1 workflow. Widespread medium to high similarity values (dark shades) indicate background contextual overlap. Task clusters are evident as dark boxes along the matrix diagonal, reflecting related actions that are naturally grouped in the sequentially ordered WBS. After applying a 0.74 cut-off threshold selected from the similarity matrix CDF, nearly all high similarity values are consolidated near the diagonal, suggesting strong potential for effective task clustering (Figure 2(B)).

Evaluation scores for cluster counts (k) ranging from 3 to 20 are shown in Figure 3. The average of the four normalised scores displays maxima at $k = 7$ and $k = 13$. Both peaks stand out prominently from neighbouring cluster counts, with similar aggregate magnitudes, making either a reasonable choice for further analysis. Neither choice produces perfect clustering of the task embeddings after applying the spectral clustering algorithm (Figure 4), as indicated by off-diagonal horizontal and vertical similarity striping in the final matrices.

This residual off-diagonal similarity likely results from imprecise action statements for each WBS element, the chosen embedding model, or non-optimal decomposition of the Design 1 workflow into discrete worksteps. At this stage, iterative refinement could improve cluster quality depending on the use case.

Figure 2 (A) Cosine similarity matrix generated for Design 1 of the joint venture analysis workflow (Holmes and Hussmo, 2025). Similarity values range from 0.0 (white) to 1.0 (black) and (B) Conditioned matrix after removing similarities below the threshold (74%) and applying min-max normalisation to the remaining values

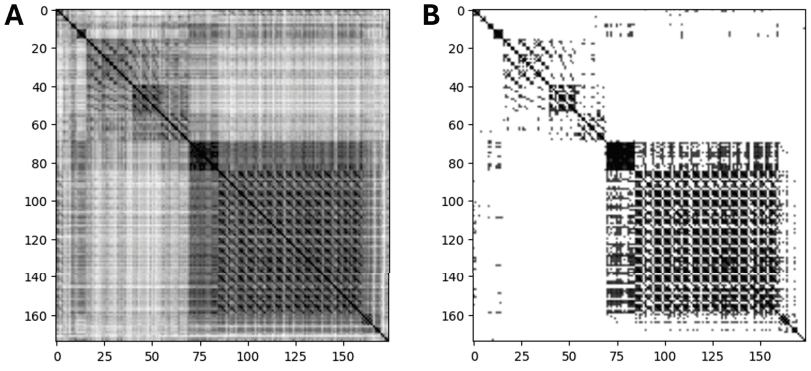
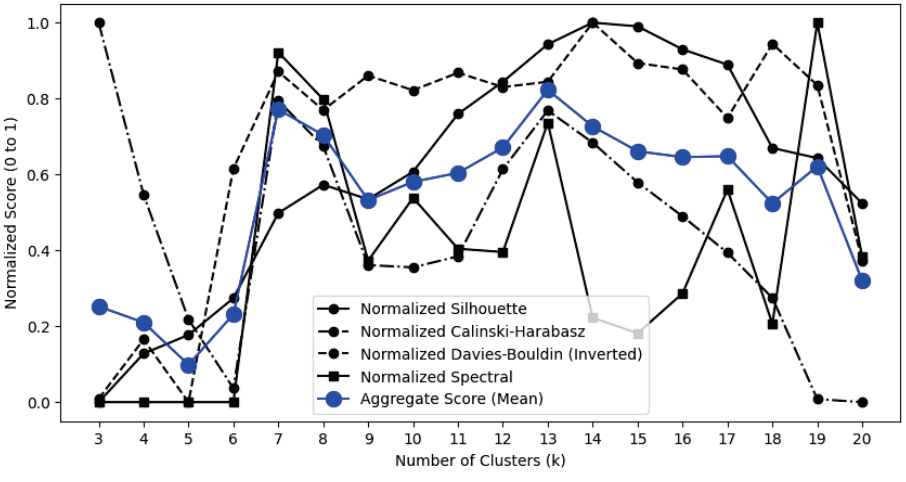


Figure 3 Individual and aggregate cluster evaluation scores for cluster counts of 3 through 20. Black lines show individual normalised metrics (Silhouette, Calinski-Harabasz, inverted Davies-Bouldin, and Spectral). The blue line shows their average, with peaks indicating recommended cluster counts (see online version for colours)



A detailed review of the task clusters reveals the roles that individual agents might have, as well as their required tools. While different cluster configurations do not yield identical task mappings, their overall consistency supports hierarchical relationships between the $k = 7$

and $k = 13$ outcomes. Figure 5 presents the $k = 7$ configuration as agents with role-based names in outer grey boxes, and agents inferred from the $k = 13$ configuration as inscribed blue boxes. The cascading lists show the tools assigned to each agent. Certain agents, such as the Schema Designer and Press Release Harvester, remain unchanged regardless of cluster count, while other agents subdivide into more specialised roles when the number of clusters increases. For example, the $k = 7$ Rubric Creator-Scorer subdivides into distinct Rubric Creator, Rubric Scorer, and Score Aggregator roles in the $k = 13$ scenario.

Figure 4 (A) Cosine similarity matrix for Design 1 after clustering with $k = 7$ and (B) the same matrix after clustering with $k = 13$

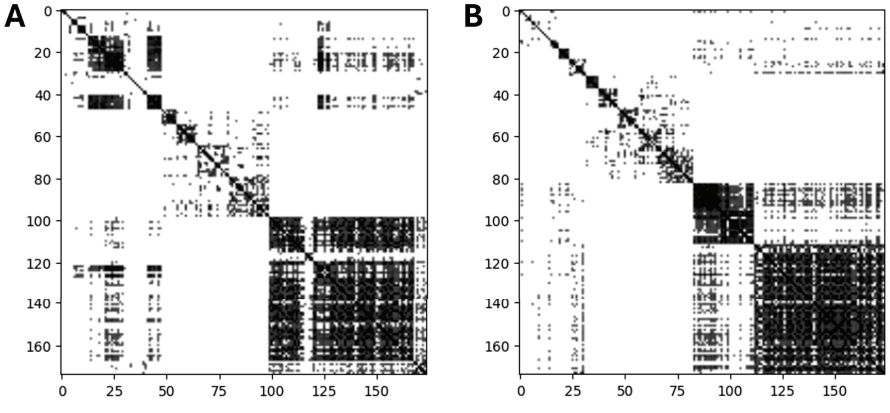
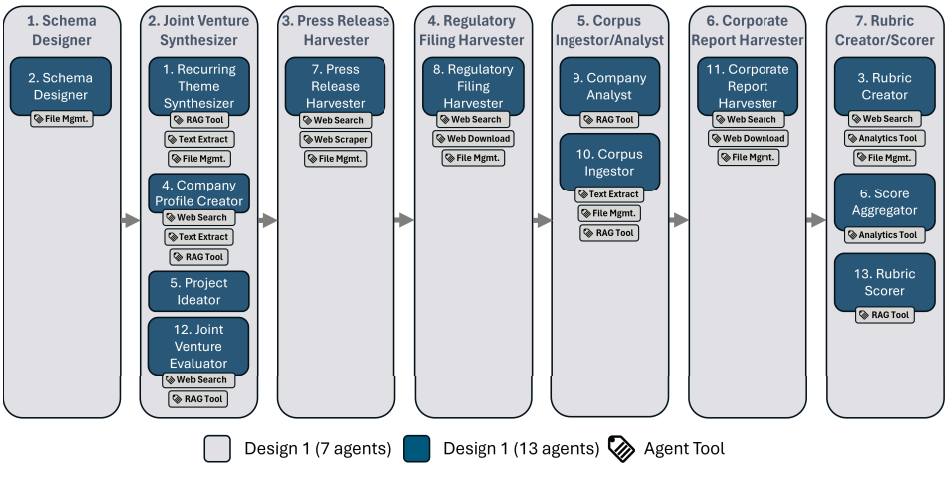


Figure 5 Comparison between agent architectures for Design 1 with $k = 7$ (grey boxes) and $k = 13$ (blue boxes). Two $k = 7$ agents (Joint Venture Synthesiser and Rubric Creator-Scorer) decompose into multiple specialised agents in the $k = 13$ configuration (see online version for colours)



These agentic relationships connect directly to agent complexity and adoptability analysis. In Figure 6(A), the AA matrix highlights agents 2 (Joint Venture Synthesiser) and 7

(Rubric Creator/Scorer) interacting with many artefacts compared to other agents in the system. Adoptability scores for these agents are correspondingly low (Figure 6(B)). Sub-scores reveal the causes: for agent 2, high tool count ($\hat{T}'$), external coupling ($\hat{E}'$), and cognitive load ($\hat{L}'$) reduce adoptability. Agent 7 imposes less of a cognitive load but has low adoptability due to tool count, lower internal cohesion (I'), and responsibility for most Design 1 worksteps ($\hat{S}'$). Both agents are natural candidates for refinement. The $k = 13$ configuration effectively distributes their responsibilities, reducing artefact load per agent (Figure 6(C)) and increasing adoptability scores (Figure 6(D)). Agents unchanged between cluster count scenarios demonstrate high adoptability scores. The Schema Designer ($k = 7$ agent 1, $k = 13$ agent 2) and Press Release Harvester ($k = 7$ agent 3, $k = 13$ agent 7) are among the top 3 agents by adoptability, with high sub-scores for cognitive load and workstep responsibilities in both cluster scenarios.

Figure 6 (A) Agent-Artefact (AA) matrix showing connectivity for Design 1 with $k = 7$. White cells = no interaction (0); grey cells = interaction (1). Agents (x-axis, numbered 1–7) match Figure 4(A) and Figure 5. Artefacts are listed along the y-axis; (B) adoptability scores and sub-scores for Design 1, $k = 7$, sorted by adoptability; (C) AA matrix for Design 1, $k = 13$ and (D) Adoptability scores for Design 1, $k = 13$ (see online version for colours)

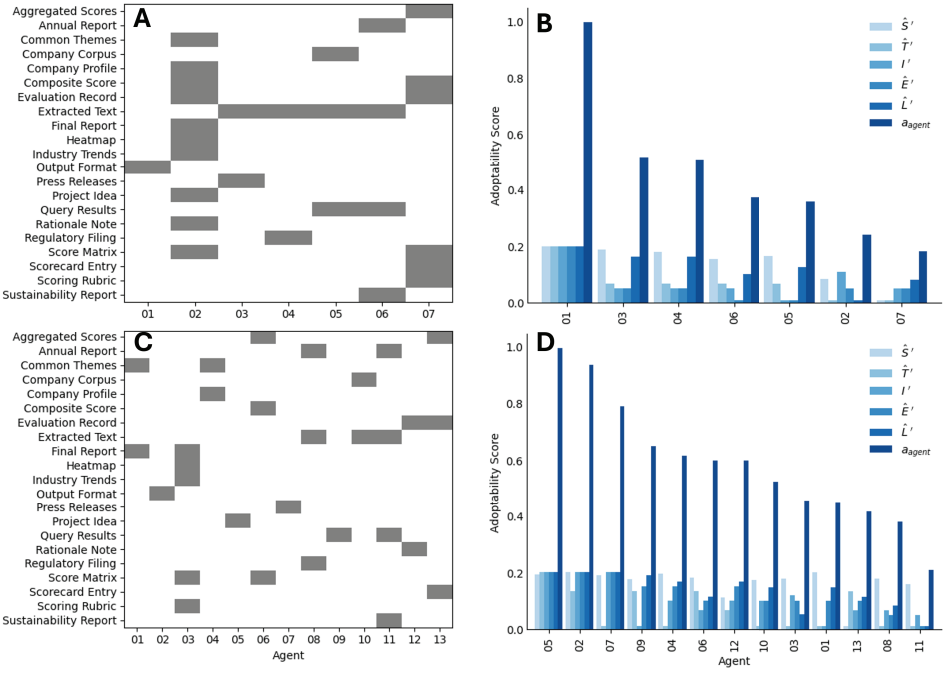


Table 2 presents the complexity and adoptability scores for each workflow design, with multiple scenarios corresponding to peaks in the cluster analysis of the respective design. Each row represents a specific agentic AI architecture for one of the workflows in Figure 5, and the agent count, agent complexity, multi-agent complexity, and adoptability scores define four dimensions to assess trade-offs between agentic AI systems. By workstep count alone, the three workflow designs grow in complexity from Design 1 to 3. The smallest

optimal cluster count also increases across designs, from $k = 7$ for Design 1 to $k = 9$ for Design 3. Within each design, agent complexity decreases with increasing number of clusters/agents. Conversely, multi-agent complexity increases as more agents are added to complete the workflow. For example, Design 3 shows that coordination complexity ($c_{multiagent}$) grows significantly with agent count; complexity more than doubles when 12 agents replace 9, then rises by another 11% when the agent count reaches 15 (Table 2).

Table 2 Complexity and adoptability scores for different designs and cluster configurations

Design	*Workflow Steps*	*Clusters/ Agents*	c_{agent}	$c_{multiagent}$	a_{system} (avg)	a_{system} (min)
1	174	7	4	20	0.46	0.18
1	174	13	3	20	0.58	0.21
2	189	8	5	16	0.56	0.23
2	189	13	5	32	0.61	0.26
3	219	9	6	8	0.63	0.22
3	219	12	4	18	0.67	0.19
3	219	15	4	20	0.68	0.27

As a multi-attribute function, adoptability exhibits interesting trends compared to complexity and step-count measures. On a full system basis, average adoptability increases consistently across Designs 1, 2, and 3. This reflects enhanced logic in the core joint venture analysis section of the workflow, as evidenced by a growing number of workflow steps and optimal agent counts (k). Although multi-agent complexity simultaneously increases, better internal cohesion, external cohesion, and agent complexity (fewer tools) effectively counterbalance this trend. Minimum adoptability scores also improve across Designs 1–3, and within each design as more agents are included in the architecture. The $k = 12$ scenario for Design 3 deviates from this trend, likely due to volatility in the clustering algorithm; the agent with the lowest a_{agent} score (Rubric Scorer) was assigned additional data analysis worksteps, negatively impacting its adoptability. Iterative refinement of WBS task wording, action statements, and embeddings may resolve this anomalous decline in adoptability with increasing k.

4 Discussion

The findings of this study offer several important insights for the design of agentic AI systems, particularly with regard to how complexity, coordination, and adoptability interconnect.

4.1 Complexity

Surprisingly, the number of tools assigned to an agent serves as a simple proxy for agent complexity. Tool count is therefore a valuable and efficient metric for early-stage design, allowing planners to verify that the final set of agents has clearly defined, focused roles and responsibilities. Furthermore, limiting agent complexity prevents the implementation of overly complex 'generalist' agents that, while capable on paper, carry a greater risk of introducing unwanted emergent behaviours in real-world applications. For example, in the $k = 7$ version of Design 1, agents 2 and 7 drove complexity score (c_{agent}) to 4 for the full

system due to their extensive tool counts (Table 2). Predictably, these two agents became the focal point for decomposition in the subsequent $k = 13$ version of Design 1 (Figure 5). Splitting the roles of these agents across multiple, more specialised agents successfully reduced role and tool complexity, as seen in the lower c_{agent} scores. These complexity scores are fundamentally about quantifying the difficulty users face in comprehending the system. Lower agent complexity means users can more easily understand the role of each agent, and agent-level metrics serve as proxies for user-level adoption barriers.

Another key insight is that increasing the total number of agents does not inherently drive higher system complexity. Instead, complexity stems from the connections and interrelations between agents. This parallels human systems, where task and knowledge differentiation do not pose an issue because '*difference without dependency is of no consequence*' (Carlile, 2002). The management of agent connections matters more than the quantity of agents or data sources involved.

For circular ecosystem formation and orchestration, this implies that expanding the quantity of R strategies (Kirchherr et al., 2017) could offer a more comprehensive perspective on circular ecosystem deployment without raising architectural complexity. Similarly, adding agents that access additional company data from internal document repositories would not increase complexity, even when such a change increases the number of data sources and provides a more holistic evaluation of joint venture creation and role distributions within the ecosystem.

4.2 Adoptability

High adoptability is associated with system configurations that use more agents, resulting in more focused task allocations, smaller toolkits, and fewer artefact interactions (Table 2). Importantly, lengthier workflows can be highly adoptable if design changes lead to more discretised and traceable tasks. Despite having more workflow steps and agents, Design 3 ($k = 15$) surpasses Design 1 ($k = 7$) in adoptability, even though they share the same agent complexity (c_{agent}) and artefact coordination complexity ($c_{multiagent}$). This underscores that intentional design focused on adoptability factors is paramount for fulfilling stakeholder needs. Consequently, architectural design analysis should precede deployment rather than simply scaling proofs of concept, as intentional design can significantly enhance the adoptability crucial for success in system value creation. For circular ecosystem formation and orchestration, this implies that all tasks required for holistic decision support must be clearly defined and incorporated into the architecture from the outset.

4.3 Role of artefacts

Artefacts are central to connecting agents and facilitating mutual understanding and interaction. However, if the number of artefacts remains fixed while the agent count increases, the dynamics of interaction can change rapidly, leading to greater coordination demands around those fixed resources. This is evident in the non-linear increase in system complexity ($c_{multiagent}$) when more clusters are used within individual designs (Table 2).

Beyond agent coordination, artefacts also enable humans to comprehend both the internal processes and ultimate outputs of agents. This resonates with the concept of *boundary objects* (Star and Griesemer, 1989), which describes how objects/artefacts enable knowledge integration and common understanding across communities (Carlile, 2002). While prior research considered human interactions with boundary objects, we observe

similar dynamics in agentic AI. Boundary object theory thus offers a useful framework for designing effective agentic architectures that foster human agency, supporting interactions across agent-agent, agent-human, and human-human contexts. This represents a promising avenue for future research.

In circular ecosystems, boundary objects produced by agents are essential for supporting human decision-making and commitment to future action. During ecosystem formation, artefacts that communicate joint venture potential enable both ideation and verification of potential partnerships. Similarly, during ecosystem orchestration, agentic AI can provide artefacts that help diverse actors align on common understanding and clarify their respective roles (Chirumalla et al., 2025).

4.4 Trustworthiness

Trustworthiness is a central issue in discussions about ethical use of AI (ECHLEGAI, 2020), encompassing both objective measures, such as accuracy and reasoning ability, and subjective human perceptions. Just as adoptability describes the degree to which the system is explicable, trustworthiness relates to a deeper level of belief. Objective metrics are important, but the subjective dimension of trust is particularly critical for decision-making use cases. Developing and refining an architecture to increase trustworthiness, however, involves associated costs. Therefore, the ultimate purpose and intended role of the agentic architecture must determine what constitutes a 'good enough' level of trustworthiness, representing a key trade-off (Hussmo, 2023).

The importance of trustworthiness varies depending on the maturity of ecosystem development efforts. In the early phases of circular ecosystem formation, trustworthiness is less critical when novelty is high and the focus is on the exploration of opportunities. However, as novelty decreases and ecosystem orchestration becomes essential, the architecture requires significantly greater levels of trustworthiness. This demonstrates the important interplay between the application context and the inherent properties of the system (Hussmo, 2023). Initial collaborative joint venture proposals, based on open-source company data, are highly consistent with the scope and purpose of ecosystem formation. Once suitable actors are identified, however, sharing private, internal datasets across actors becomes a key strategy for enhancing the objective trustworthiness of the architecture and its application.

5 Conclusions

This study demonstrates how agentic AI architectures can support the formation and orchestration of circular ecosystems through a framework for system design and pre-deployment assessment. A central insight is that agent complexity, quantified by the number of tools assigned to each agent, serves as a practical proxy for early-stage design decisions. By decomposing generalist agents into more specialised roles, system complexity can be managed without sacrificing functionality. System complexity also appears to be driven less by the number of agents and more by the interconnections and dependencies among them. This underscores the importance of intentional architectural design, where differentiation without dependency does not increase complexity.

System adoptability is critical to achieving higher user acceptance, driving the value creation sought by AI system development. Our analysis shows that architectural designs

that use more agents, better task distribution, and simpler toolkits resulted in significantly higher adoptability, with an increase of up to 50% in scores across three workflow prototypes (see Table 2). This suggests that intentional design prioritising transparency, traceability, and human comprehensibility should precede system deployment, rather than simply scaling proofs-of-concept that demonstrate technical capabilities. Moreover, artefacts serve as important boundary objects, promoting mutual understanding, alignment, and effective decision-making among agents and stakeholders within the ecosystem. Ultimately, trustworthiness in complex agentic systems must be actively managed, especially as systems mature from exploratory tools into core technologies supporting ecosystem decision-making.

Future research should prioritise several key directions. First, real-world case studies are needed to validate and empirically quantify the impact of these architectural principles on complex decision-making and ecosystem formation and orchestration. Second, enhancing the integrity and accuracy of system inputs by gaining access to reliable internal firm data is vital for credibility, although this integration must be handled with care to ensure performance gains do not compromise company competitiveness. Third, this approach can be applied to model real-world dynamics, enabling more efficient responses to dynamic shifts in geopolitics and evolving stakeholder objectives. Ultimately, human verification is necessary to assess the subjective trustworthiness of the system and ensure all outputs are actionable through effective human-systems interactions. These findings support agile formation, reformation, and orchestration of circular ecosystems during crises, a topic previously identified as critical in the literature (Aryee et al., 2025).

By systematically addressing these design and research opportunities, agentic AI can more effectively move from theory to practical use in circular ecosystem formation and orchestration. This progress can add valuable perspectives to multi-agent systems literature and promote the development of more sustainable, collaborative, and resilient sociotechnical practices.

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Conflict of interest

The authors declare that they have no conflicts of interest.

Originality and responsibility

This manuscript is the original work of the authors and has not been published or submitted elsewhere. All interpretations, analysis, and conclusions are solely those of the authors.

Use of AI Tools

Generative AI was an integral part of the proposed architectural analysis workflow, as detailed in Section 2.5. AI tools also helped verify grammar and improve readability during manuscript preparation without altering the original ideas or scientific conclusions of the authors. The authors reviewed and approved all content and assume full responsibility for its accuracy and integrity.

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